

Unit 1. Data Diagnostics and Problem Framing

Business Data Analytics: Introductory Machine Learning

Why this unit has no algorithms

After half a semester of data wrangling, the next step is modeling. This unit defers it. The most consequential errors in applied analysis are made before any algorithm is fit:

- an identifier column treated as a numeric feature;
- a feature recorded after the event being predicted;
- 99% accuracy on a 99%-imbalanced dataset;
- a random split applied to time-series data.

Each yields strong in-sample scores and unreliable behavior in deployment.

This unit develops a seven-question diagnostic checklist; every subsequent unit returns to it.

A note on vocabulary

The terms above (*feature*, *accuracy*, *train/test split*) are introduced formally in definition boxes as they are needed.

1 The Diagnostic Checklist

Seven questions for every dataset, posed before any model is fit. Each is unpacked in a dedicated section; subsequent units revisit the list whenever a new algorithm is introduced.

- Q1. Column types.** What is each column: numeric, categorical, ordinal, datetime, or identifier?
- Q2. Distributions, gaps, outliers.** How is each variable distributed? What is missing, and is the missingness itself signal?
- Q3. Provenance and timing.** How was this data generated, and *when* was each field written?
- Q4. Target definition.** What exactly is y ? When can it be observed?
- Q5. Prediction or explanation.** Are we forecasting y , or asking what drives it?
- Q6. Leakage.** Does any feature carry information from *after* y is observed?
- Q7. Metric.** Does the score we report reflect the actual cost of being wrong?

On the convex-analysis primer

§9 closes with a primer on convex functions and minimization. It is not an eighth question but the mathematical vocabulary on which subsequent units depend (“minimize what loss subject to what constraints”). Intended for one reading and subsequent reference.

2 From Data Wrangling to Modeling

The first half of the semester answered descriptive questions: counts, means, joins, summaries. Modeling addresses a different question: given previously unseen inputs, what is the output? This section establishes the required vocabulary.

2.1 The data layout

A modeling dataset retains the rectangular form, but its columns now play asymmetric roles: each is a **feature** (input), the **target** (output), or neither (an identifier).

Definition 1.1. For a dataset with rows indexed by $i = 1, \dots, n$:

- the i -th row is one **observation**;

- a column used as model input is a **feature** (or *predictor*); features for observation i form a vector $\mathbf{x}_i \in \mathbb{R}^p$;
- the column the model is asked to predict is the **target** (*response, outcome*); call its value y_i ;
- identifier columns (customer ID, row number) are neither.

The tibble below has $n = 4$ observations, $p = 3$ features, one target, and one identifier:

```
1 library(tidyverse)
2 customers <- tibble(
3   customer_id = c(1001, 1002, 1003, 1004), # identifier (not a feature)
4   age         = c(34, 52, 28, 41),         # feature
5   region      = c("N", "S", "N", "E"),     # feature
6   spent_usd   = c(120, 340, 75, 210),     # feature
7   churned     = c(0, 1, 0, 1)             # target
8 )
9 customers
```

```
## # A tibble: 4 x 5
```

```
##   customer_id  age region spent_usd churned
```

##	<dbl>	<dbl>	<chr>	<dbl>	<dbl>
## 1	1001	34	N	120	0
## 2	1002	52	S	340	1
## 3	1003	28	N	75	0
## 4	1004	41	E	210	1

2.2 Training set and test set

A model that perfectly fits its training data is uninformative about new observations. To detect such memorization, a portion of the dataset is held out and never inspected during training.

Definition 1.2. The **training set** is the subset of rows used to estimate the model. The **test set** is a disjoint subset, locked away during training, used only once at the end to estimate how the model behaves on previously unseen rows.

Test-set protocol

The test set is consulted *exactly once*, at the end, for the final score. Any inspection during training invalidates the experiment. Unit 4 generalizes this discipline to cross-validation.

2.3 The modeling pipeline: model, fit, predict

A *model* is a family of candidate functions; *fitting* selects one from the family using the training data; *prediction* evaluates that function on new feature values.

Definition 1.3. A **model** is a parameterized family of functions $\{f_\theta : \mathbb{R}^p \rightarrow \mathcal{Y} \mid \theta \in \Theta\}$, where θ is a parameter vector and $\mathcal{Y}$ is the space the target lives in. To **fit** the model is to choose an estimate $\hat{\theta}$ from the training data; the chosen function $f_{\hat{\theta}}$ is the **fitted model**. To **predict** for a new feature vector $\mathbf{x}_\star$ is to compute $\hat{y}_\star = f_{\hat{\theta}}(\mathbf{x}_\star)$.

$$\{(\mathbf{x}_i, y_i)\}_{i=1}^{n_{\text{train}}} \xrightarrow{\text{fit}} \hat{\theta} \xrightarrow{f_{\hat{\theta}}(\cdot)} \hat{y}_\star \text{ for any new } \mathbf{x}_\star.$$

Every algorithm in this course (linear regression, logistic regression, decision trees, support vector machines, K-means) is a specific choice of the family $\{f_\theta\}$ together with a specific procedure for selecting $\hat{\theta}$.

2.4 What “good” means

A model is judged not by its fit on training data but by its predictive accuracy on previously unseen observations. This property is called **generalization**.

Definition 1.4. A model **generalizes well** if its predictions on previously unseen feature vectors are close (in some specified sense) to the true target values.

Why “in some specified sense”

“Close” depends on the metric, and the metric depends on the cost of being wrong. That is Q7 of the checklist; §8 returns to it.

3 Systematic EDA Workflow

Questions **Q1** (column types) and **Q2** (distributions, missingness, outliers) are answered first; every subsequent question depends on them. *Exploratory data analysis* (EDA) is the reproducible procedure that does so: a small ordered set of operations applied to every dataset, not a dashboard glance.

3.1 The five column types

Definition 1.5. Every column in a modeling dataset belongs to one of five types:

- **Numeric** (continuous or count): values where arithmetic is meaningful (**age**, **spent_usd**).
- **Categorical** (nominal): unordered labels (**region** $\in \{N, S, E, W\}$).
- **Ordinal**: ordered labels with no meaningful numeric distance (**satisfaction** $\in \{\text{low}, \text{mid}, \text{high}\}$).
- **Datetime**: timestamps with explicit time semantics.
- **Identifier**: row keys; treat as metadata, never as a feature.

The most common feature-engineering error in this course is treating an identifier as a numeric feature. R issues no warning; the resulting model implicitly assigns customer 10001 twice the magnitude of customer 5000 on whatever scale the column happens to occupy. `glimpse()`, which prints each column's type and a few sample values, is the first defence.

```
1 glimpse(customers)
```

```
## Rows: 4
```

```
## Columns: 5
```

```
## $ customer_id <dbl> 1001, 1002, 1003, 1004
## $ age          <dbl> 34, 52, 28, 41
## $ region       <chr> "N", "S", "N", "E"
## $ spent_usd    <dbl> 120, 340, 75, 210
## $ churned      <dbl> 0, 1, 0, 1
```

3.2 Inspecting distributions

For **numeric** columns: histogram, boxplot, or density estimate. Diagnose skewness, multimodality, and spikes at zero or at boundary values, which often flag data-collection artefacts.

For **categorical** columns: a bar count of level frequencies. Diagnose severe class imbalance and rare levels that lack sufficient support for learning.

Why distributions matter even before any model

A column that is 99% zeros; a “rating” on a 1–5 scale that occasionally registers 99; a Gaussian-looking variable with a cluster at -9999 . These are not algorithm-specific issues but properties of the data that no subsequent choice of method can repair if undetected at this stage.

The **diamonds** dataset (bundled with **ggplot2**) provides sufficient sample size to make distributional

shape visible:

```
1 library(ggplot2)
2 ggplot(diamonds, aes(price)) + geom_histogram(bins = 40)
```

```
1 ggplot(diamonds, aes(cut)) + geom_bar()
```

3.3 Missingness as signal

A missing value is itself an observation: the fact of unobservation carries information about the data-generating process.

Definition 1.6. Three patterns of missingness, in increasing severity:

- **MCAR** (missing completely at random): missingness is independent of every variable, observed or not.
- **MAR** (missing at random): missingness depends on observed variables but not on the missing value itself.
- **MNAR** (missing not at random): missingness depends on the value that is missing.

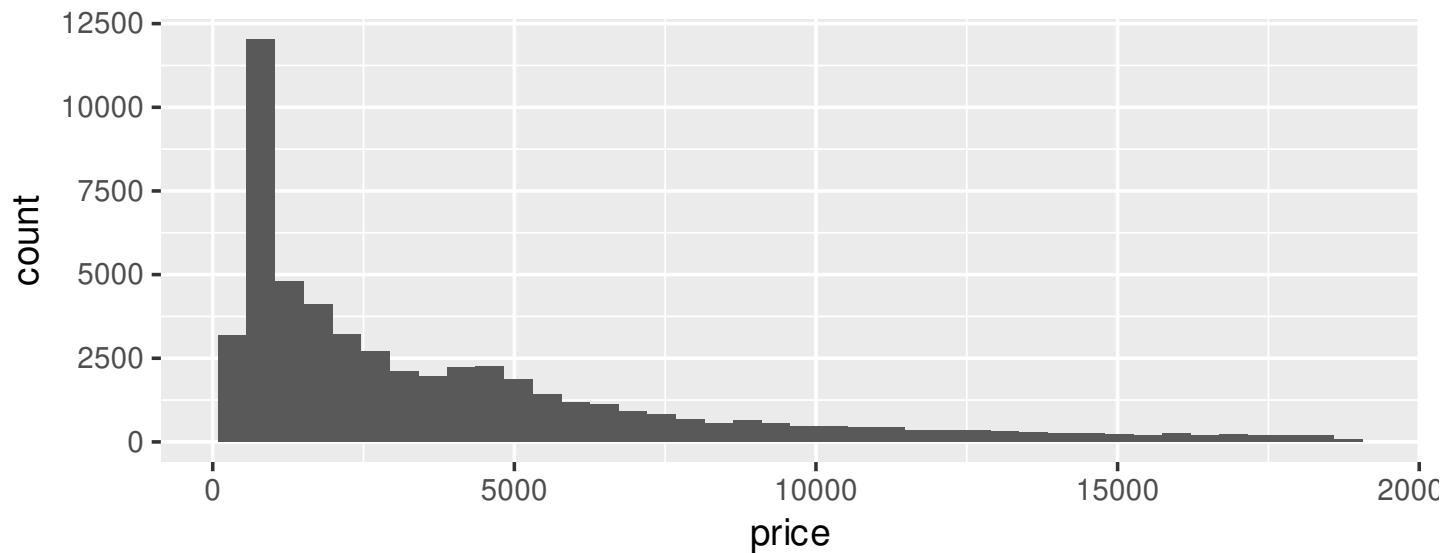


Figure 1.1: Histogram of diamond **price**: heavy right tail and a spike near zero are immediately visible.

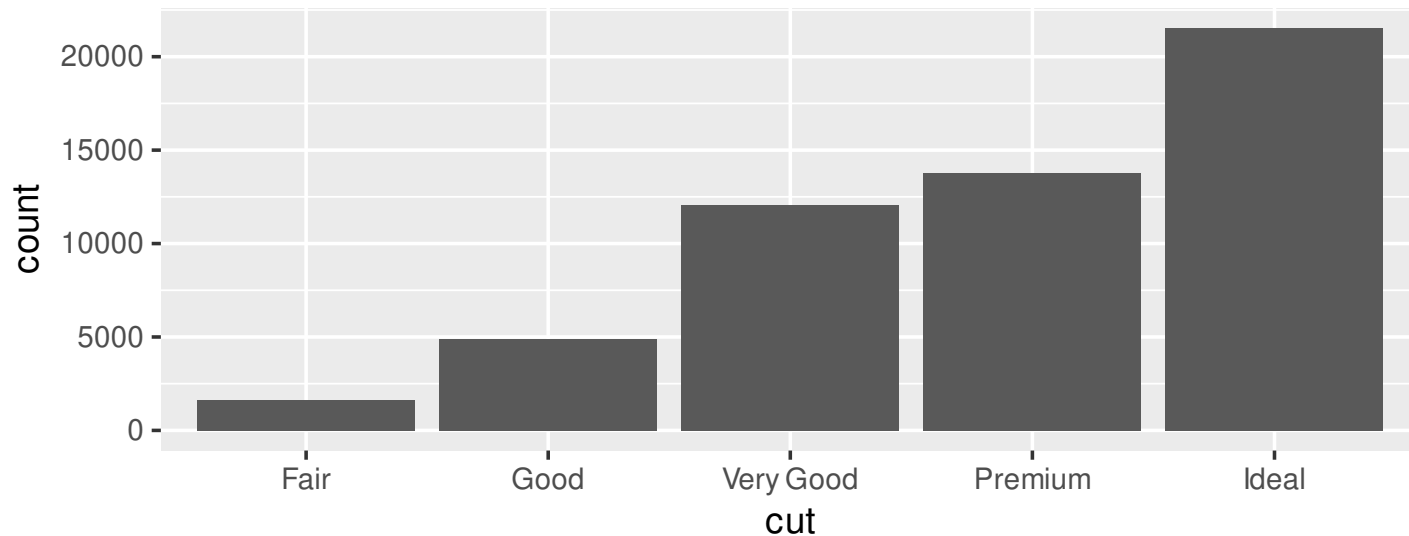


Figure 1.2: Bar count of `cut` quality levels: **Ideal** dominates; the rare **Fair** level may not have enough rows to learn from.

Imputation is not free

Mean imputation is the most common preprocessing default. It is unbiased only under MCAR; under MAR it biases coefficient estimates; under MNAR it both biases the estimates and discards the signal carried by the missingness pattern. The recommended default is to encode the missingness indicator before imputing (cf. §10).

The minimum diagnostic for any dataset is a per-column NA tally:

```
1 airquality |>
2   summarise(across(everything(), ~ sum(is.na(.)))) |>
3   pivot_longer(everything(), names_to = "column", values_to = "n_missing") |>
4   arrange(desc(n_missing))
```

```
## # A tibble: 6 x 2
##   column  n_missing
##   <chr>    <int>
## 1 Ozone      37
## 2 Solar.R     7
## 3 Wind        0
## 4 Temp        0
```

## 5 Month	0
## 6 Day	0

3.4 Outliers

Definition 1.7. A point is a **statistical outlier** of a numeric column if it lies far from the bulk of the values (e.g. outside $\bar{x} \pm 3s$, or outside the Tukey range $[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]$). A point is a **domain outlier** if its value is implausible given what the variable is supposed to represent.

The two definitions diverge frequently. `age = 200` is statistically extreme and a domain error. `income = $2,000,000` is statistically extreme yet legitimate; deletion would remove signal.

Outlier protocol

Each outlier should be investigated and classified (legitimate, data error, or unusual but real) before any retain / transform / winsorize / remove decision. Routine deletion beyond 3σ is identified as a pitfall in §10.

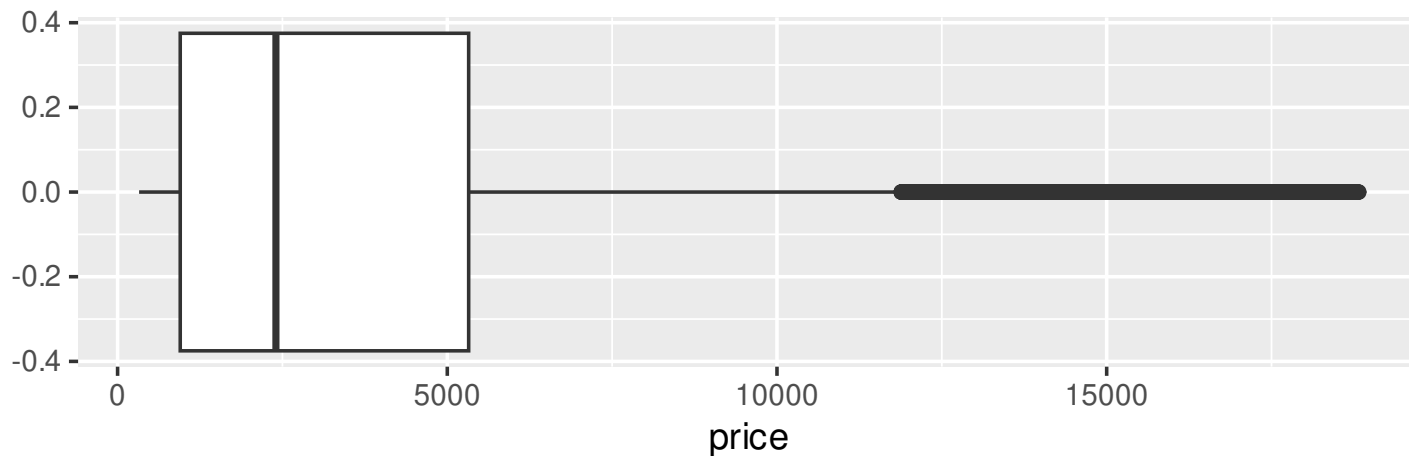


Figure 1.3: Boxplot of diamond **price**: the long upper whisker and dense outlier points reflect a real, heavy-tailed distribution, not data errors.

```
1 ggplot(diamonds, aes(price)) + geom_boxplot()
```

3.5 A minimal EDA toolkit

A reusable starting script for any new `df`:

```
1 # 1. Structure: types and sample values for each column.
2 glimpse(df)
3
4 # 2. Per-column summary: range, central tendency, NAs.
5 skimr::skim(df)
6
7 # 3. Distributions of every numeric column.
8 df |> select(where(is.numeric)) |>
9   pivot_longer(everything()) |>
10  ggplot(aes(value)) + geom_histogram(bins = 40) +
11  facet_wrap(~ name, scales = "free")
12
13 # 4. Distributions of every categorical column.
14 df |> select(where(is.character) | where(is.factor)) |>
15   pivot_longer(everything()) |>
16   ggplot(aes(value)) + geom_bar() +
17   facet_wrap(~ name, scales = "free") + coord_flip()
18
```

```
19 # 5. Missingness pattern, sorted.  
20 df |> summarise(across(everything(), ~ sum(is.na(.)))) |>  
21   pivot_longer(everything(), names_to = "column", values_to = "n_missing") |>  
22   arrange(desc(n_missing))
```

Together, these five blocks answer **Q1** and **Q2** for any dataset in roughly a minute, and should be run before any modeling decision.

4 Data Provenance and Temporal Structure

Question **Q3** concerns how each column was generated and when it was written. Most leakage and time-series errors in this course trace back to incorrect answers here.

4.1 Three questions about every column

Before treating any column as a feature, answer:

1. **Who wrote it?** A human typist, an automated system, an upstream model's output, a manual override?
2. **When was it written?** At the moment of the event, at end-of-day batch, at query time?

3. **When does it become readable to a downstream system?** (Often *not* the same as when it was written: there is usually a delay.)

A column whose three answers are unknown is unusable as a feature; either the answers are obtained, or the column is excluded.

4.2 The three temporal axes

Definition 1.8. For each row of a dataset, three time stamps are conceptually distinct:

- **Event time** t_{ev} : when the underlying real-world event happened.
- **Record time** t_{rec} : when the value was actually written to the system.
- **Prediction time** $t_{\star}$: when the model is asked to make its prediction.

In an idealized source these timestamps coincide (a sensor that writes immediately and is queried instantly). In real systems they diverge:

- A customer cancels at t_{ev} ; the cancellation is logged at $t_{\text{rec}} = t_{\text{ev}} + 3$ days after manual review.
- A monthly aggregate (**total_purchases**) is written at end-of-month, but its t_{ev} covers the entire month; it is a *summary* of past events, not an event.

4.3 The freshness gap and joins

Definition 1.9. The **freshness gap** at prediction time $t_\star$ is $t_\star - t_{\text{rec}}$: how stale the most recently available value is. Zero gap means instant writes; positive gap is the delay before a new value reaches the model.

A common error is to ignore the gap and join a current-state snapshot onto historical events, attaching values that did not yet exist at t_{ev} :

```
1 events <- tibble(  
2   customer_id = c(1, 1, 1, 2, 2),  
3   event_time  = as.Date(c("2026-01-01", "2026-02-01", "2026-03-01",  
4                        "2026-01-15", "2026-03-10")),  
5   amount      = c(120, 240, 360, 90, 180)  
6 )  
7 state_now <- tibble(  
8   customer_id      = c(1, 2),  
9   total_purchases = c(720, 270), # cumulative as of "now" (post-2026-03-10)  
10  segment          = c("VIP", "regular") # current segment, not historical  
11 )
```

```
12 events |> left_join(state_now, by = "customer_id")
```

```
## # A tibble: 5 x 5
##   customer_id event_time amount total_purchases segment
##         <dbl> <date>      <dbl>          <dbl> <chr>
## 1             1 2026-01-01      120            720 VIP
## 2             1 2026-02-01      240            720 VIP
## 3             1 2026-03-01      360            720 VIP
## 4             2 2026-01-15        90            270 regular
## 5             2 2026-03-10      180            270 regular
```

Every row in the joined table now carries **total_purchases** and **segment** values computed after its t_{ev} . The January row for customer 1 holds her March cumulative spend. A model trained on this table performs exceptionally on the historical data and fails in deployment.

Join constraint

Every feature value joined onto an event must satisfy $t_{\text{rec}} \leq t_{\text{ev}}$. When in doubt, annotate each column with its t_{rec} and reject any join that violates the constraint. Unit 4 generalizes the rule to cross-validation; Unit 7 revisits it for non-stationary segments.

5 Target Definition

Question **Q4** concerns y itself: its formal definition and the time at which it becomes observable. Imprecise answers propagate into every metric and report, yet target definition is the step most often skipped in practice.

5.1 What exactly is y ?

Definition 1.10. A **target definition** specifies, in writing:

- the value space $\mathcal{Y}$ (binary, multiclass, count, real-valued, time-to-event, ...);
- the unit of observation (per customer? per transaction? per session?);
- the operational rule that turns raw events into a single value of y for that unit.

A single business term can encompass incompatible target definitions. “Churn” may mean an explicit cancellation, the absence of a purchase in 90 days, or cancellation within 30 days of a renewal: three distinct y ’s, datasets, and models. The choice must be documented before any modeling.

```
1 events <- tibble(  
2   customer_id = c(1, 1, 2, 2, 3),  
3   event       = c("purchase", "purchase", "purchase", "cancel", "purchase"),
```

```
4   event_time = as.Date(c("2026-01-15", "2026-02-20", "2025-08-10",
5                           "2026-01-05", "2026-01-01"))
6 )
7 t_star <- as.Date("2026-04-25")
8
9 # (a) Cancel-event churn.
10 churn_a <- events |> group_by(customer_id) |>
11   summarise(churn = any(event == "cancel"), .groups = "drop")
12
13 # (b) Inactivity churn: no purchase in the 90 days before t_star.
14 churn_b <- events |> filter(event == "purchase") |>
15   group_by(customer_id) |>
16   summarise(churn = (t_star - max(event_time)) > 90, .groups = "drop")
17
18 inner_join(churn_a, churn_b, by = "customer_id",
19            suffix = c("_a", "_b"))
```

```
## # A tibble: 3 x 3
##   customer_id churn_a churn_b
##         <dbl> <lgl>   <lgl>
## 1           1 FALSE    FALSE
```

## 2	2 TRUE	TRUE
## 3	3 FALSE	TRUE

The two definitions disagree on customer 3: no cancellation event, yet no purchase in the past 90 days. Both rules are defensible; exactly one is the operative y .

5.2 When can y be observed?

Definition 1.11. The **observation lag** of y is the delay between the event of interest and the moment its y value is reliably known. A row is usable for training only if its y is already observed by the time we are training.

Examples of substantial lag: a default is confirmed 90 days after a missed payment; a customer is labeled churned only after a 90-day inactivity window; a marketing conversion is counted at the end of the attribution window.

Training-row constraint

Each training row $(\mathbf{x}_i, y_i)$ must satisfy $t_{\text{rec}}(y_i) \leq t_*$: the label is observable at the simulated prediction time. Rows whose y is not yet known belong in a separate pending set, never in

training.

5.3 Survivorship and selection on y

Selection bias on the target

The dataset typically contains only rows for which y is observable. Mutual-fund histories exclude closed funds; satisfaction surveys exclude non-respondents; loan-repayment tables exclude rejected applicants. The fitted model is implicitly a model of conditional y given survival; if the deployment population includes non-survivors, predictions are biased.

Diagnostic test: can the rows missing from the dataset due to unobservable y be described in one sentence? If not, the dataset is not ready for modeling.

6 Prediction or Explanation

Question **Q5** concerns the type of question being answered. *Predicting y* and *explaining y* are easily conflated in ordinary usage; they are distinct objectives, and confusing them yields models that are either confidently misleading or quietly inert.

Definition 1.12. Two distinct problems:

- A **prediction problem** asks: given $\mathbf{x}$, what is the best estimate of y ? Judged by error on unseen rows. Any predictive feature is welcome, even if its causal role is unknown.
- An **explanation problem** asks: which inputs *cause* y to move, and by how much? Judged against the true causal mechanism. A feature that merely correlates with the cause is misleading.

Example 1.1. Ice-cream sales predict drowning deaths well, both rising in summer. A prediction model accepts ice-cream sales as a useful regressor for drowning risk. An explanation model rejects the inference: banning ice cream prevents no drownings; the common cause is hot weather. Identical data, identical correlation, divergent verdicts.

Where this course stands

This course is **prediction-first**: algorithms are judged by out-of-sample prediction quality. The question “why does this model predict well?” is permitted as a debugging tool; “what causes y ?” is not formally answered. Causal inference (do-calculus, instrumental variables, randomized

experiments) is a separate course; genuinely causal questions such as “will this discount increase sales?” lie beyond prediction alone.

AI tools and the distinction

A general-purpose AI assistant, asked which features matter for y , will report whatever variables a regression flags as significant, as though that answered a causal question. It does not. A large coefficient is a property of the fit; causation requires assumptions the model cannot verify. AI-generated explanatory claims should be read with this distinction in mind.

7 Target Leakage

Question **Q6** is the central concern of this unit. Target leakage is the most common cause of strong validation followed by deployment failure, and the hardest to detect because its symptoms resemble success.

7.1 What is target leakage?

Definition 1.13. A feature x_j **leaks the target** if its value at training time depends on information that, at the moment we are pretending to predict, would not yet have been available. The leak need not be the value of y itself: any quantity that is causally downstream of y , or co-emitted with y , qualifies.

The characteristic warning sign is a single feature that nearly determines y . Such a feature appears to be an excellent predictor and is, in most cases, y in disguise.

7.2 Three forms of leakage

Definition 1.14. The three patterns to watch for:

- **Post-event features:** a column whose value is written only after y occurs (e.g. “support tickets in the cancellation week” for a churn model).
- **Train–test contamination:** information from the test rows leaks into preprocessing applied to the training rows (standardising using the global mean, target-encoding using all rows, feature selection on the full table).

- **Group leakage:** the same entity appears in both the train and the test split, so the model memorizes entities rather than generalizing to new ones.

7.3 Detecting leakage

The diagnostic is primarily logical, with one numerical safeguard:

1. For every feature column, can you certify $t_{\text{rec}} \leq t_{\star}$ (§4)? If unsure, drop the column until proven safe.
2. For every preprocessing step (centering, imputation, encoding, feature selection), confirm that it is computed using *only* training rows, never on the full table before splitting.
3. For every train/test split, verify that no entity (customer, patient, store) appears in both sides.
4. Numeric tripwire: if any *single* feature classifies near-perfectly, treat it as a leak suspect.

A toy demonstration. `x_leak` is constructed as `y` plus a small perturbation; `x_real` is independent of `y`:

```
1 set.seed(1)
2 n <- 1000
```

```
3 y      <- rbinom(n, 1, 0.5)
4 x_leak <- y + rnorm(n, 0, 0.1)  # almost equal to y
5 x_real <- rnorm(n)              # independent of y
6
7 accuracy <- function(pred, true) mean(pred == true)
8 c(leak = accuracy(x_leak > 0.5, y),
9   real = accuracy(x_real > 0,   y))
```

```
## leak real
## 1.000 0.499
```

An accuracy of 0.99 from a single feature is the diagnostic signal. In a clean dataset, no individual column should classify that well; in a churn or default problem, such a column is almost certainly a transformation of y .

7.4 Companion lab

The companion lab supplies a churn dataset that initially reports 99% accuracy. The feature `support_time` is populated only after cancellations are processed, a textbook post-event leak. The exercise is to identify this without prompting. The corrected pipeline reports approximately 78%, the realistic figure.

Pedagogical rationale

Reading about leakage is far less effective than mistaking a 99% score for genuine performance and then learning why it is not. As a working principle, no high-score result should be trusted until an attempt has been made to falsify it.

8 Evaluation Metrics: a First Look

Question **Q7** asks whether the headline number reflects the underlying objective. Accuracy is both the obvious starting point and the obvious failure case. ROC, PR curves, threshold tuning, and cost-sensitive scoring receive full treatment in Unit 3; this section installs only the minimum vocabulary.

8.1 Accuracy

Definition 1.15. For a classifier producing predictions $\hat{y}_i$ on n test rows, **accuracy** is

$$\text{Acc} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\hat{y}_i = y_i\},$$

the fraction of predictions that match the truth.

Accuracy is straightforward to compute and report, and equally straightforward to be misled by. Subsequent units return to its limitations.

8.2 The confusion matrix

Definition 1.16. For a binary classifier with positive class “1” and negative class “0”, the **confusion matrix** on a test set is the 2×2 table

	$\hat{y} = 1$	$\hat{y} = 0$
$y = 1$	TP	FN
$y = 0$	FP	TN

where TP = true positives, FP = false positives, FN = false negatives, TN = true negatives. Note

that $\text{Acc} = (\text{TP} + \text{TN})/n$.

A single accuracy figure collapses these four counts into one, discarding information: two classifiers with markedly different TP/FP/FN/TN profiles can register identical accuracy.

8.3 Why accuracy fails on imbalance

If 99% of test rows are negative, the trivial classifier that always predicts negative achieves 99% accuracy while detecting no positives. For business problems in which the rare class is the relevant one (fraud, default, churn, conversion), this is the dominant failure mode of accuracy:

```
1 set.seed(1)
2 n      <- 1000
3 y      <- rbinom(n, 1, 0.01)      # 1% positives
4 y_hat  <- rep(0, n)              # trivial: always predict 0
5
6 mean(y_hat == y)                 # accuracy
```

```
## [1] 0.986
```

```
1 table(predicted = y_hat, actual = y)
```

```
##          actual
## predicted    0    1
##          0 986  14
```

Accuracy is near 0.99, positive recall is zero, and the model is uninformative for the question of interest.

8.4 The cost view

In nearly every business setting, the four corners of the confusion matrix carry distinct costs. A missed fraud (FN) may be 50× more costly than a falsely flagged clean transaction (FP). The appropriate scalar objective is a weighted combination of these counts, and the appropriate score-to-decision threshold depends on those weights.

Unit 3 framework

Unit 3 introduces precision, recall, F_1 , ROC and PR curves, calibration, and threshold tuning, and integrates a business cost matrix directly into model selection. For Unit 1 the operative point is that accuracy is the default, the default fails on imbalance, and the Unit 3 machinery exists to remedy it.

9 Convex Analysis Prerequisites

This section is not a diagnostic question but the mathematical vocabulary on which the remainder of the course depends. Every supervised algorithm covered (linear regression, logistic regression, Ridge, Lasso, SVMs) is defined by a loss function minimized over a parameter set. Three terms are required: *loss*, *minimization*, and *convex*.

9.1 Loss functions and empirical risk

Definition 1.17. A **loss function** $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}$ measures how wrong a single prediction $\hat{y}$ is relative to the truth y . For training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ and a model family $\{f_\theta\}$, the **empirical risk** (or **average training loss**) is

$$L(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_\theta(\mathbf{x}_i)).$$

Three loss functions you will meet in the next three units:

- **Squared loss** (Unit 2, regression): $\ell(y, \hat{y}) = (y - \hat{y})^2$.

- **Zero–one loss** (background): $\ell(y, \hat{y}) = \mathbb{1}\{y \neq \hat{y}\}$. Easy to interpret; intractable to optimise.
- **Logistic loss** (Unit 3, classification): $\ell(y, \hat{y}) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$ for $y \in \{0, 1\}$, $\hat{y} \in (0, 1)$.

9.2 The minimization problem

Fitting a model now has a precise meaning: choose $\hat{\theta}$ that minimizes empirical risk over the allowed parameter set,

$$\hat{\theta} = \arg \min_{\theta \in \Theta} L(\theta).$$

Algorithms differ in their choice of ℓ , of $\{f_{\theta}\}$, and of Θ (e.g. $\Theta = \mathbb{R}^p$ for OLS, $\Theta = \{\theta : \|\theta\|_1 \leq t\}$ for Lasso). Tractability of the minimization depends almost entirely on a single property of L : *convexity*.

9.3 Convex functions

Definition 1.18. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if its domain $\text{dom } f$ is a convex set and, for every $\mathbf{x}, \mathbf{y} \in \text{dom } f$ and every $\theta \in [0, 1]$,

$$f(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta) f(\mathbf{y}).$$

It is **strictly convex** if the inequality is strict whenever $\mathbf{x} \neq \mathbf{y}$ and $\theta \in (0, 1)$; **concave** if $-f$ is convex.

Figure 1.4 illustrates the inequality: the chord between any two points on the graph lies on or above the graph, with strict inequality except at the endpoints in the strictly convex case.

Convex / concave functions on $\mathbb{R}$.

- Convex: affine $ax + b$; exponential e^{ax} ; power x^α on $x > 0$ for $\alpha \geq 1$ or $\alpha \leq 0$; $|x|^\alpha$ for $\alpha \geq 1$; positive part $\max\{x, 0\}$ (the ReLU activation).
- Concave: affine; x^α on $x > 0$ for $0 \leq \alpha \leq 1$; $\log x$; entropy $-x \log x$ on $x > 0$.

Convex functions on $\mathbb{R}^n$.

- affine: $\mathbf{a}^\top \mathbf{x} + b$;
- every norm $\|\cdot\|$ (any ℓ_p -norm, $\|\cdot\|_\infty$, spectral norm of a matrix, ...);
- sum of squares $\|\mathbf{x}\|_2^2 = \sum_i x_i^2$;

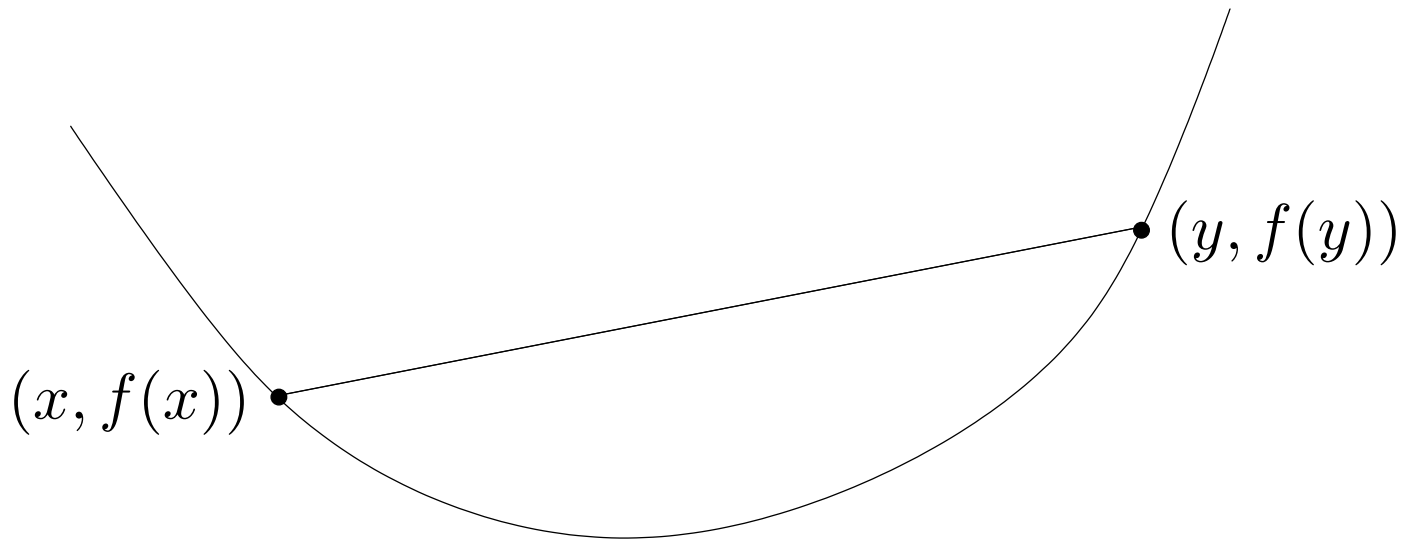


Figure 1.4: Convexity in one dimension: the chord between any two graph points lies on or above the graph itself.

- $\max(\mathbf{x}) = \max_i x_i$;
- log-sum-exp $\log \sum_i e^{x_i}$ (a smooth convex relaxation of max; reappears in Unit 3 as the negative log-likelihood of multinomial logistic regression).

Theorem 1.19 (Norms are convex). Every norm $\|\cdot\|$ on $\mathbb{R}^n$ is a convex function.

Proof of Theorem 1.19

Recall that a norm satisfies the triangle inequality $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ and absolute homogeneity $\|\alpha \mathbf{u}\| = |\alpha| \|\mathbf{u}\|$. Let $\theta \in [0, 1]$ and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Apply the triangle inequality to $\mathbf{u} = \theta \mathbf{x}$, $\mathbf{v} = (1 - \theta) \mathbf{y}$ and use homogeneity:

$$\|\theta \mathbf{x} + (1 - \theta) \mathbf{y}\| \leq \|\theta \mathbf{x}\| + \|(1 - \theta) \mathbf{y}\| = \theta \|\mathbf{x}\| + (1 - \theta) \|\mathbf{y}\|.$$

That is exactly the convexity inequality applied to $f(\mathbf{x}) = \|\mathbf{x}\|$. ■

This is the most-used convexity result in Unit 2: the Lasso penalty $\|\boldsymbol{\beta}\|_1$ and the Ridge penalty $\|\boldsymbol{\beta}\|_2^2$ are convex, and a nonnegative combination with a convex loss preserves convexity (Theorem 1.31).

9.4 Gradient, Hessian, PSD: a small toolkit

The constructions below require a brief vocabulary from multivariate calculus and matrix analysis.

Definition 1.20 (Gradient and Hessian). For a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **gradient** at $\mathbf{x}$ is the column vector of partial derivatives,

$$\nabla f(\mathbf{x}) = (\partial f / \partial x_1, \dots, \partial f / \partial x_n)^\top \in \mathbb{R}^n.$$

If f is twice differentiable, the **Hessian** at $\mathbf{x}$ is the symmetric $n \times n$ matrix of second partial derivatives,

$$[\nabla^2 f(\mathbf{x})]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}).$$

Chain rule along a line. For $\mathbf{x} \in \text{dom } f$, $\mathbf{v} \in \mathbb{R}^n$, and the one-variable restriction $g(t) = f(\mathbf{x} + t\mathbf{v})$, the multivariate chain rule gives

$$g'(t) = \nabla f(\mathbf{x} + t\mathbf{v})^\top \mathbf{v}, \quad g''(t) = \mathbf{v}^\top \nabla^2 f(\mathbf{x} + t\mathbf{v}) \mathbf{v}.$$

At $t = 0$, $g'(0) = \nabla f(\mathbf{x})^\top \mathbf{v}$ is the **directional derivative** of f at $\mathbf{x}$ in direction $\mathbf{v}$. These two identities will be exploited in the restriction-to-a-line arguments developed below.

Definition 1.21 (Positive (semi-)definite). A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (PSD), denoted $\mathbf{A} \succcurlyeq 0$, if

$$\mathbf{v}^\top \mathbf{A} \mathbf{v} \geq 0 \quad \forall \mathbf{v} \in \mathbb{R}^n.$$

It is **positive definite** (PD), denoted $\mathbf{A} \succ 0$, if the inequality is strict for every $\mathbf{v} \neq \mathbf{0}$. The notation $\mathbf{A} \succcurlyeq \mathbf{B}$ (resp. $\mathbf{A} \succ \mathbf{B}$) means $\mathbf{A} - \mathbf{B}$ is PSD (resp. PD); the relation $\succcurlyeq$ is a partial order on the symmetric matrices, the **Loewner order**.

9.5 Restriction to a line

Theorem 1.22 (Restriction to a line). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if and only if, for every $\mathbf{x} \in \text{dom } f$ and every direction $\mathbf{v} \in \mathbb{R}^n$, the one-variable function

$$g(t) = f(\mathbf{x} + t\mathbf{v}), \quad \text{dom } g = \{t \in \mathbb{R} : \mathbf{x} + t\mathbf{v} \in \text{dom } f\},$$

is convex in t .

This is the standard reduction: multivariate convexity is verified one direction at a time. The multivariate question becomes one-dimensional, where Taylor expansions and second-derivative tests apply directly.

Proof of Theorem 1.22

($\Rightarrow$) Assume f is convex. Fix $\mathbf{x}, \mathbf{v}$. For $t_1, t_2 \in \text{dom } g$ and $\theta \in [0, 1]$,

$$g((1 - \theta)t_1 + \theta t_2) = f((1 - \theta)(\mathbf{x} + t_1\mathbf{v}) + \theta(\mathbf{x} + t_2\mathbf{v})) \leq (1 - \theta)g(t_1) + \theta g(t_2),$$

where the inequality is the convexity of f . Hence g is convex.

($\Leftarrow$) Assume g is convex for every $(\mathbf{x}, \mathbf{v})$. Given any $\mathbf{x}, \mathbf{y} \in \text{dom } f$, set $\mathbf{v} = \mathbf{y} - \mathbf{x}$, so $g(0) = f(\mathbf{x})$ and $g(1) = f(\mathbf{y})$. For $\theta \in [0, 1]$,

$$f((1 - \theta)\mathbf{x} + \theta\mathbf{y}) = g(\theta) \leq (1 - \theta)g(0) + \theta g(1) = (1 - \theta)f(\mathbf{x}) + \theta f(\mathbf{y}). \quad \blacksquare$$

Example 1.2. (*Quadratic forms.*) Let $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{P} \mathbf{x}$ with $\mathbf{P}$ symmetric. Restrict to the line $\mathbf{x} + t\mathbf{v}$:

$$g(t) = \mathbf{x}^\top \mathbf{P} \mathbf{x} + 2t \mathbf{v}^\top \mathbf{P} \mathbf{x} + t^2 \mathbf{v}^\top \mathbf{P} \mathbf{v}.$$

This is a one-dimensional quadratic in t with leading coefficient $\mathbf{v}^\top \mathbf{P} \mathbf{v}$. It is convex in t for every $\mathbf{v}$ iff $\mathbf{v}^\top \mathbf{P} \mathbf{v} \geq 0$ for every $\mathbf{v}$, i.e., iff $\mathbf{P} \succcurlyeq 0$. By Theorem 1.22, f is convex on $\mathbb{R}^n$ iff $\mathbf{P} \succcurlyeq 0$.

Example 1.3. (*Sum of squares.*) For $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $\mathbf{y} \in \mathbb{R}^n$, the residual sum of squares

$$f(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$$

is convex in $\boldsymbol{\beta}$. Restricting to the line $\boldsymbol{\beta} + t\mathbf{v}$,

$$g(t) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 - 2t \mathbf{v}^\top \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + t^2 \|\mathbf{X}\mathbf{v}\|_2^2,$$

a 1-D quadratic with leading coefficient $\|\mathbf{X}\mathbf{v}\|_2^2 \geq 0$. Convexity holds for every $\mathbf{v}$, so by Theorem 1.22, f is convex on $\mathbb{R}^p$ regardless of $\mathbf{X}$; this is the Unit 2 OLS objective, established without computing a $p \times p$ Hessian.

Example 1.4. (*Negative log-determinant on the positive-definite cone.*) Let $f(\mathbf{X}) = -\log \det \mathbf{X}$ with $\text{dom } f = \mathbf{S}_{++}^n$ (symmetric positive-definite matrices). Pick $\mathbf{X} \succ 0$ and a symmetric direction $\mathbf{V}$. Restricting to $\mathbf{X} + t\mathbf{V}$, let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $\mathbf{X}^{-1/2}\mathbf{V}\mathbf{X}^{-1/2}$. Using $\mathbf{X} + t\mathbf{V} = \mathbf{X}^{1/2}(\mathbf{I} + t\mathbf{X}^{-1/2}\mathbf{V}\mathbf{X}^{-1/2})\mathbf{X}^{1/2}$,

$$g(t) = -\log \det(\mathbf{X} + t\mathbf{V}) = -\log \det \mathbf{X} - \sum_{i=1}^n \log(1 + t\lambda_i).$$

Each summand $t \mapsto -\log(1 + t\lambda_i)$ has second derivative $\lambda_i^2/(1 + t\lambda_i)^2 \geq 0$ wherever defined, hence is convex; g is a sum of convex functions, hence convex. By Theorem 1.22, f is convex on $\mathbf{S}_{++}^n$. This is exactly the negative Gaussian log-likelihood for the precision matrix; the same argument is the convexity backbone of the graphical Lasso.

Use in statistics

Restriction to a line is the standard proof tool when the multivariate Hessian is opaque, typical of Gaussian likelihoods, the graphical Lasso, and covariance estimation. The convexity question

reduces to a one-dimensional second derivative.

9.6 First-order condition

Theorem 1.23 (First-order condition). Let f be differentiable on the open convex set $\text{dom } f$. Then f is convex if and only if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}), \quad \forall \mathbf{x}, \mathbf{y} \in \text{dom } f.$$

The first-order Taylor approximation of f at any point is therefore a **global underestimator** of f ; see Figure 1.5.

Proof of Theorem 1.23

($\Rightarrow$) Assume f is convex. Fix $\mathbf{x}, \mathbf{y}$ and let $g(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$. By Theorem 1.22, g is convex on $[0, 1]$, with $g(0) = f(\mathbf{x})$, $g(1) = f(\mathbf{y})$, and $g'(0) = \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$. For $t \in (0, 1]$, convexity of g gives $g(t) \leq (1-t)g(0) + tg(1)$, i.e. $\frac{g(t)-g(0)}{t} \leq g(1) - g(0)$. Letting $t \rightarrow 0^+$ yields $g'(0) \leq g(1) - g(0)$, which substitutes back to $\nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x})$.

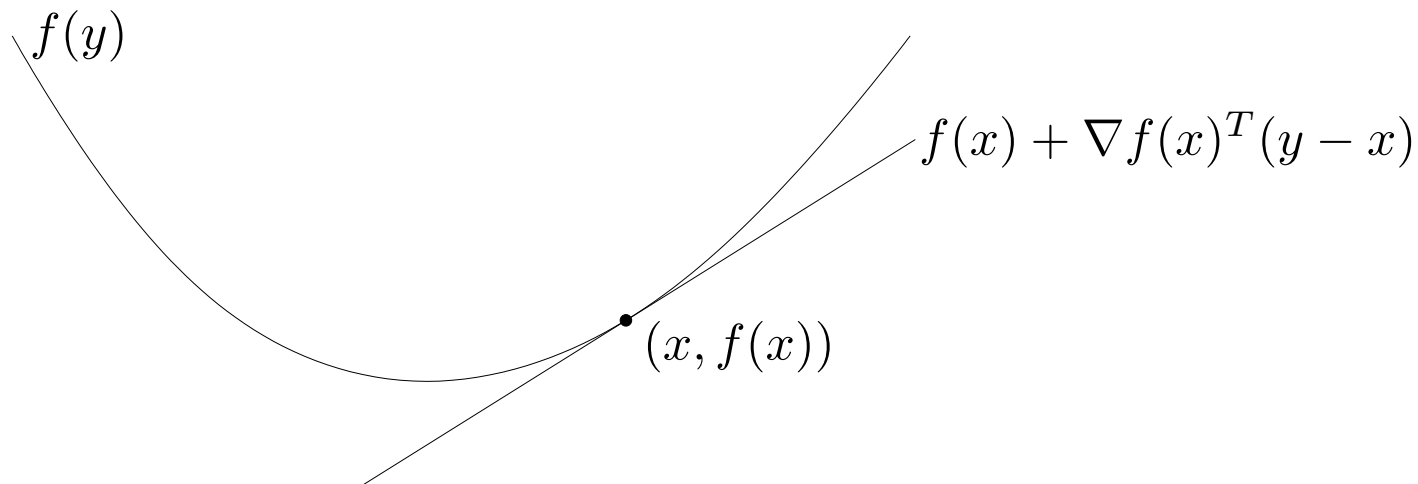


Figure 1.5: First-order Taylor approximation (dashed line) as a global underestimator of a convex function.

($\Leftarrow$) Assume the inequality. Given $\mathbf{x}, \mathbf{y}$ and $\theta \in [0, 1]$, set $\mathbf{z} = \theta\mathbf{x} + (1 - \theta)\mathbf{y}$. Apply the assumed inequality at $(\mathbf{z}, \mathbf{x})$ and $(\mathbf{z}, \mathbf{y})$:

$$f(\mathbf{x}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^\top (\mathbf{x} - \mathbf{z}), \quad f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^\top (\mathbf{y} - \mathbf{z}).$$

Multiply the first by θ , the second by $1 - \theta$, and add. The gradient terms collapse to $\nabla f(\mathbf{z})^\top (\theta\mathbf{x} + (1 - \theta)\mathbf{y} - \mathbf{z}) = 0$, leaving $\theta f(\mathbf{x}) + (1 - \theta)f(\mathbf{y}) \geq f(\mathbf{z})$. ■

Optimality corollary

If $\mathbf{x}^*$ is interior to $\text{dom } f$ and $\nabla f(\mathbf{x}^*) = \mathbf{0}$, then $f(\mathbf{y}) \geq f(\mathbf{x}^*)$ for every $\mathbf{y}$: a stationary point is automatically a global minimum. This justifies the “set gradient to zero, solve” procedure used in Unit 2 (Ridge, Lasso) and Unit 3 (logistic regression). For non-differentiable cases such as the Lasso penalty $\|\boldsymbol{\beta}\|_1$, the subgradient extension below replaces $\nabla f(\mathbf{x})$ with a subgradient $\mathbf{g} \in \partial f(\mathbf{x})$.

Definition 1.24 (Subgradient and subdifferential). For a convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{x} \in \text{dom } f$, a vector $\mathbf{g} \in \mathbb{R}^n$ is a **subgradient** of f at $\mathbf{x}$ if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{g}^\top(\mathbf{y} - \mathbf{x}), \quad \forall \mathbf{y} \in \text{dom } f.$$

The set of all such $\mathbf{g}$ is the **subdifferential** $\partial f(\mathbf{x})$. Existence of subgradients at interior points is established in Theorem 1.29.

When f is differentiable at $\mathbf{x}$, $\partial f(\mathbf{x}) = \{\nabla f(\mathbf{x})\}$ and the subgradient inequality reduces to Theorem 1.23. At a non-differentiable point the subdifferential is a non-trivial closed convex set: for example, $\partial|x|$ at $x = 0$ equals the interval $[-1, 1]$. Unit 2 develops the subdifferential of $\|\boldsymbol{\beta}\|_1$ and applies Definition 1.24 in place of Theorem 1.23 to derive the soft-thresholding update for the Lasso.

9.7 Epigraph and sublevel sets

Definition 1.25. For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$:

- the **epigraph** is $\text{epi } f = \{(\mathbf{x}, t) \in \mathbb{R}^{n+1} : \mathbf{x} \in \text{dom } f, f(\mathbf{x}) \leq t\}$, the region on or above the graph;
- the **α -sublevel set** is $C_\alpha = \{\mathbf{x} \in \text{dom } f : f(\mathbf{x}) \leq \alpha\}$.

Theorem 1.26 (Epigraph characterization). f is convex $\iff \text{epi } f$ is a convex set in $\mathbb{R}^{n+1}$. If f is convex, every sublevel set C_α is convex (the converse is false). See Figure 1.6.

Proof of Theorem 1.26

($\Rightarrow$) Let f be convex and take $(\mathbf{x}_1, t_1), (\mathbf{x}_2, t_2) \in \text{epi } f$, so $f(\mathbf{x}_i) \leq t_i$. For $\theta \in [0, 1]$,

$$f(\theta\mathbf{x}_1 + (1 - \theta)\mathbf{x}_2) \leq \theta f(\mathbf{x}_1) + (1 - \theta)f(\mathbf{x}_2) \leq \theta t_1 + (1 - \theta)t_2.$$

Hence $(\theta\mathbf{x}_1 + (1 - \theta)\mathbf{x}_2, \theta t_1 + (1 - \theta)t_2) \in \text{epi } f$, so $\text{epi } f$ is convex.

($\Leftarrow$) Let $\text{epi } f$ be convex. Take $(\mathbf{x}_1, f(\mathbf{x}_1))$ and $(\mathbf{x}_2, f(\mathbf{x}_2))$ in $\text{epi } f$. By convexity of $\text{epi } f$, $(\theta\mathbf{x}_1 + (1 - \theta)\mathbf{x}_2, \theta f(\mathbf{x}_1) + (1 - \theta)f(\mathbf{x}_2)) \in \text{epi } f$, so $f(\theta\mathbf{x}_1 + (1 - \theta)\mathbf{x}_2) \leq \theta f(\mathbf{x}_1) + (1 - \theta)f(\mathbf{x}_2)$.

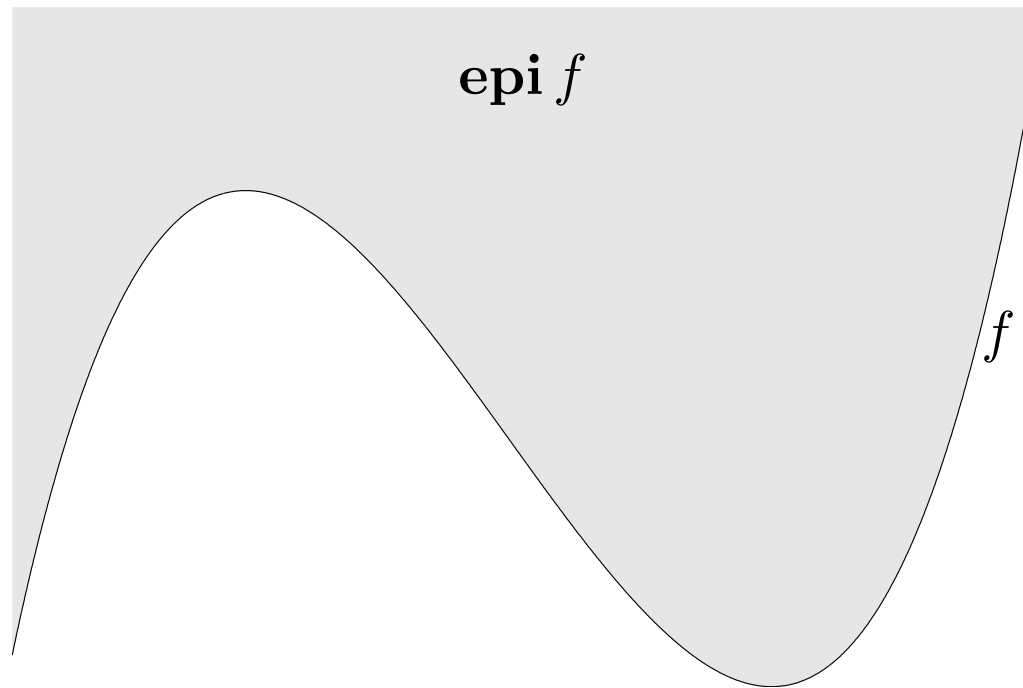


Figure 1.6: Epigraph of f : the shaded region on or above the graph.



Functions and sets

Theorem 1.26 is the formal correspondence between convex functions and convex sets: every theorem on convex sets yields a theorem on convex functions, and conversely. Unit 6 builds the maximum-margin SVM directly on this correspondence: the SVM primal is a constrained convex problem whose feasible region is an intersection of halfspaces.

9.8 Supporting hyperplane and subgradient existence

The non-emptiness asserted in Definition 1.24 is the geometric fact that a convex set has a tangent hyperplane at every boundary point. We prove this and derive subgradient existence as a corollary. The projection step in the proof rests on one algebraic identity for the squared Euclidean norm:

Lemma 1.27 (Strict convexity of $\|\cdot\|^2$). For every $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and $\theta \in [0, 1]$,

$$\|(1 - \theta)\mathbf{u} + \theta\mathbf{v}\|^2 = (1 - \theta)\|\mathbf{u}\|^2 + \theta\|\mathbf{v}\|^2 - \theta(1 - \theta)\|\mathbf{u} - \mathbf{v}\|^2.$$

In particular, $\|\cdot\|^2$ is *strictly* convex on $\mathbb{R}^n$: the value of $\|(1 - \theta)\mathbf{u} + \theta\mathbf{v}\|^2$ lies strictly below the

chord $(1 - \theta)\|\mathbf{u}\|^2 + \theta\|\mathbf{v}\|^2$ whenever $\theta \in (0, 1)$ and $\mathbf{u} \neq \mathbf{v}$.

Proof of Lemma 1.27

Expanding $\|\mathbf{w}\|^2 = \mathbf{w}^\top \mathbf{w}$ gives

$$\|(1 - \theta)\mathbf{u} + \theta\mathbf{v}\|^2 = (1 - \theta)^2\|\mathbf{u}\|^2 + 2\theta(1 - \theta)\mathbf{u}^\top \mathbf{v} + \theta^2\|\mathbf{v}\|^2.$$

Subtracting from the chord and using $(1 - \theta) - (1 - \theta)^2 = \theta(1 - \theta) = \theta - \theta^2$,

$$\begin{aligned} (1 - \theta)\|\mathbf{u}\|^2 + \theta\|\mathbf{v}\|^2 - \|(1 - \theta)\mathbf{u} + \theta\mathbf{v}\|^2 \\ = \theta(1 - \theta)[\|\mathbf{u}\|^2 - 2\mathbf{u}^\top \mathbf{v} + \|\mathbf{v}\|^2] = \theta(1 - \theta)\|\mathbf{u} - \mathbf{v}\|^2, \end{aligned}$$

which rearranges to the stated identity. The right-hand side is non-negative and vanishes only when $\theta(1 - \theta) = 0$ (i.e. $\theta \in \{0, 1\}$) or $\mathbf{u} = \mathbf{v}$, so the chord lies strictly above otherwise. ■

Theorem 1.28 (Supporting hyperplane). Let $C \subseteq \mathbb{R}^n$ be a non-empty convex set and $\mathbf{x}_0$ a boundary point of C . There exists $\mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ such that

$$\mathbf{a}^\top \mathbf{x} \leq \mathbf{a}^\top \mathbf{x}_0, \quad \forall \mathbf{x} \in C.$$

The hyperplane $\{\mathbf{x} : \mathbf{a}^\top \mathbf{x} = \mathbf{a}^\top \mathbf{x}_0\}$ is a **supporting hyperplane** to C at $\mathbf{x}_0$.

The proof goes through three steps: a projection lemma, strict separation of a point from a closed convex set, and a boundary limit.

Proof of Theorem 1.28

Step 1: Projection lemma. If $K \subseteq \mathbb{R}^n$ is closed convex and $\mathbf{z} \notin K$, there exists a unique $\mathbf{x}^* \in K$ minimizing $\|\mathbf{z} - \mathbf{x}\|$, and

$$(\mathbf{z} - \mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \leq 0, \quad \forall \mathbf{x} \in K.$$

The map $\mathbf{x} \mapsto \|\mathbf{z} - \mathbf{x}\|^2$ is continuous and tends to infinity, so its restriction to $K \cap \{\|\mathbf{x}\| \leq R\}$ for large R attains a minimum on this compact set; let $\mathbf{x}^* \in K$ be a minimizer with minimum

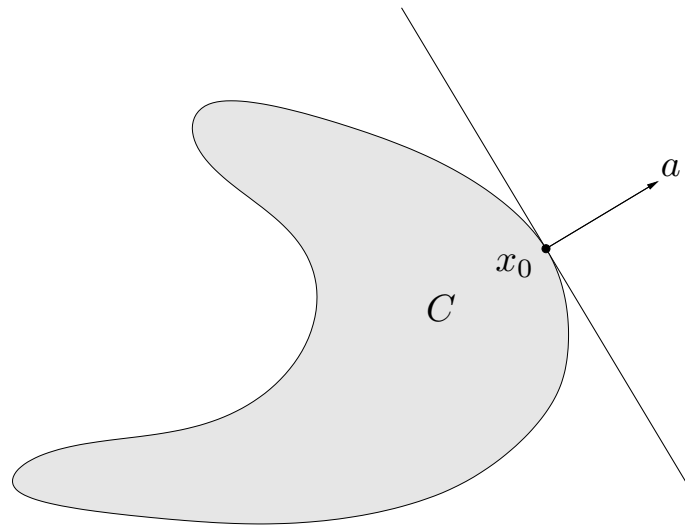


Figure 1.7: Supporting hyperplane to a convex set C at the boundary point $\mathbf{x}_0$: the hyperplane passes through $\mathbf{x}_0$ and C lies on one side.

value $d^2 = \|\mathbf{z} - \mathbf{x}^*\|^2$. For uniqueness, suppose $\tilde{\mathbf{x}} \in K$ satisfies $\|\mathbf{z} - \tilde{\mathbf{x}}\|^2 = d^2$ with $\tilde{\mathbf{x}} \neq \mathbf{x}^*$. Apply Lemma 1.27 with $\mathbf{u} = \mathbf{z} - \mathbf{x}^*$, $\mathbf{v} = \mathbf{z} - \tilde{\mathbf{x}}$, $\theta = 1/2$:

$$\left\| \mathbf{z} - \frac{1}{2}(\mathbf{x}^* + \tilde{\mathbf{x}}) \right\|^2 = \frac{1}{2}d^2 + \frac{1}{2}d^2 - \frac{1}{4}\|\mathbf{x}^* - \tilde{\mathbf{x}}\|^2 < d^2.$$

The midpoint $\frac{1}{2}(\mathbf{x}^* + \tilde{\mathbf{x}}) \in K$ by convexity, contradicting the minimality of d^2 ; hence $\mathbf{x}^*$ is unique. For the variational inequality, fix any $\mathbf{x} \in K$; for $t \in [0, 1]$, the segment $\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*)$ lies in K by convexity, so

$$\|\mathbf{z} - \mathbf{x}^* - t(\mathbf{x} - \mathbf{x}^*)\|^2 \geq \|\mathbf{z} - \mathbf{x}^*\|^2.$$

Expanding and differentiating in t at $t = 0^+$ gives $-2(\mathbf{z} - \mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \geq 0$, the claimed inequality.

Step 2: Strict separation when $\mathbf{d} \notin \overline{C}$. Apply Step 1 with $K = \overline{C}$ and $\mathbf{z} = \mathbf{d}$. Set $\mathbf{a} = \mathbf{d} - \mathbf{x}^* \neq \mathbf{0}$ and $b = \mathbf{a}^\top \mathbf{x}^*$. For every $\mathbf{x} \in \overline{C}$, $\mathbf{a}^\top (\mathbf{x} - \mathbf{x}^*) \leq 0$, i.e. $\mathbf{a}^\top \mathbf{x} \leq b$; meanwhile $\mathbf{a}^\top \mathbf{d} = b + \|\mathbf{a}\|^2 > b$.

Step 3: Boundary limit. Let $\mathbf{x}_0$ be a boundary point of C . Choose $\mathbf{d}_n \notin \overline{C}$ with $\mathbf{d}_n \rightarrow \mathbf{x}_0$; for each n , Step 2 yields $\mathbf{a}_n \neq \mathbf{0}$ with $\mathbf{a}_n^\top \mathbf{x} \leq \mathbf{a}_n^\top \mathbf{d}_n$ for all $\mathbf{x} \in C$. Normalize so $\|\mathbf{a}_n\| = 1$. The unit sphere is compact, so a subsequence $\mathbf{a}_{n_k}$ converges to some $\mathbf{a}$ with $\|\mathbf{a}\| = 1$. Passing to the

limit in $\mathbf{a}_{n_k}^\top \mathbf{x} \leq \mathbf{a}_{n_k}^\top \mathbf{d}_{n_k}$ gives $\mathbf{a}^\top \mathbf{x} \leq \mathbf{a}^\top \mathbf{x}_0$ for all $\mathbf{x} \in C$. ■

Theorem 1.29 (Existence of subgradients). For convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and any $\mathbf{x}_0$ in the interior of $\text{dom } f$, $\partial f(\mathbf{x}_0)$ is non-empty.

Proof of Theorem 1.29

The epigraph $\text{epi } f = \{(\mathbf{x}, t) : f(\mathbf{x}) \leq t\} \subseteq \mathbb{R}^{n+1}$ is convex (Theorem 1.26), and $(\mathbf{x}_0, f(\mathbf{x}_0))$ is a boundary point: any neighbourhood contains points $(\mathbf{x}_0, f(\mathbf{x}_0) - \varepsilon) \notin \text{epi } f$. Apply Theorem 1.28 at this boundary point: there exists $(\mathbf{a}, \beta) \in \mathbb{R}^{n+1} \setminus \{\mathbf{0}\}$ with

$$\mathbf{a}^\top \mathbf{x} + \beta t \leq \mathbf{a}^\top \mathbf{x}_0 + \beta f(\mathbf{x}_0), \quad \forall (\mathbf{x}, t) \in \text{epi } f.$$

For any $\mathbf{x} \in \text{dom } f$, letting $t \rightarrow +\infty$ on the left forces $\beta \leq 0$. If $\beta = 0$, the inequality reduces to $\mathbf{a}^\top \mathbf{x} \leq \mathbf{a}^\top \mathbf{x}_0$ for every $\mathbf{x} \in \text{dom } f$; but $\mathbf{x}_0$ is interior, so taking $\mathbf{x} = \mathbf{x}_0 \pm \varepsilon \mathbf{a}$ for small $\varepsilon > 0$ gives $\pm \varepsilon \|\mathbf{a}\|^2 \leq 0$, i.e. $\mathbf{a} = \mathbf{0}$, contradicting $(\mathbf{a}, \beta) \neq \mathbf{0}$. Hence $\beta < 0$; rescale by $1/|\beta|$ so that $\beta = -1$.

Setting $t = f(\mathbf{x})$,

$$\mathbf{a}^\top \mathbf{x} - f(\mathbf{x}) \leq \mathbf{a}^\top \mathbf{x}_0 - f(\mathbf{x}_0) \iff f(\mathbf{x}) \geq f(\mathbf{x}_0) + \mathbf{a}^\top (\mathbf{x} - \mathbf{x}_0),$$

so $\mathbf{a} \in \partial f(\mathbf{x}_0)$. ■

Use in this course

Theorem 1.29 is invoked twice: once in Unit 2, where the subdifferential of $\|\boldsymbol{\beta}\|_1$ drives the soft-thresholding update for Lasso, and once in the proof of Jensen's inequality (Theorem 1.32) below. Theorem 1.28 itself reappears in Unit 6 (SVM), where the maximum-margin classifier is constructed by separating two convex sets of points by a hyperplane.

9.9 Second-order condition and examples

Theorem 1.30 (Second-order condition). Let f be twice continuously differentiable on the open convex set $\text{dom } f$. Then

- f is convex $\iff \nabla^2 f(\mathbf{x}) \succcurlyeq 0$ for every $\mathbf{x} \in \text{dom } f$;
- $\nabla^2 f(\mathbf{x}) \succ 0$ for every $\mathbf{x} \Rightarrow f$ is strictly convex.

Proof of Theorem 1.30

By Theorem 1.22, f is convex iff $g(t) = f(\mathbf{x} + t\mathbf{v})$ is convex in t for every $\mathbf{x} \in \text{dom } f$, $\mathbf{v} \in \mathbb{R}^n$. Twice differentiability gives

$$g''(t) = \mathbf{v}^\top \nabla^2 f(\mathbf{x} + t\mathbf{v}) \mathbf{v}.$$

The multivariate question reduces to a one-dimensional fact: *a twice-differentiable g on an interval is convex iff $g''(t) \geq 0$ throughout.*

For the 1-D fact: by Taylor's theorem with Lagrange remainder, for any a, b in the interval,

$\theta \in [0, 1]$, and $z = \theta a + (1 - \theta)b$ there exist ξ_a, ξ_b such that

$$g(a) = g(z) + g'(z)(a - z) + \frac{1}{2} g''(\xi_a)(a - z)^2, \quad g(b) = g(z) + g'(z)(b - z) + \frac{1}{2} g''(\xi_b)(b - z)^2.$$

Form $\theta g(a) + (1 - \theta)g(b)$ and use $\theta(a - z) + (1 - \theta)(b - z) = 0$ to cancel the linear term:

$$\theta g(a) + (1 - \theta)g(b) = g(z) + \frac{1}{2} [\theta g''(\xi_a)(a - z)^2 + (1 - \theta) g''(\xi_b)(b - z)^2].$$

The bracket is ≥ 0 iff $g'' \geq 0$ wherever evaluated, giving $g(z) \leq \theta g(a) + (1 - \theta)g(b)$ iff g is convex.

Translating back: $g''(t) \geq 0$ for every $\mathbf{v}$ and every admissible t is exactly $\mathbf{v}^\top \nabla^2 f(\mathbf{y}) \mathbf{v} \geq 0$ for every $\mathbf{v}$ and every $\mathbf{y} \in \text{dom } f$, i.e. $\nabla^2 f \succcurlyeq 0$ on $\text{dom } f$. The strict version replaces $\geq$ with $>$ and convex with strictly convex throughout. ■

Examples.

- **Quadratic** $f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{P} \mathbf{x} + \mathbf{q}^\top \mathbf{x} + r$: $\nabla^2 f = \mathbf{P}$. Convex $\iff \mathbf{P} \succcurlyeq 0$; concave $\iff \mathbf{P} \preccurlyeq 0$.

- **Least-squares** $f(\boldsymbol{\beta}) = \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2$: $\nabla^2 f = 2\mathbf{X}^\top \mathbf{X} \succcurlyeq 0$, so f is convex for *every* $\mathbf{X}$; strictly convex iff $\mathbf{X}$ has full column rank. (Unit 2 OLS lives here.)
- **Quadratic-over-linear** $f(x, y) = x^2/y$ on $\{y > 0\}$:

$$\nabla^2 f(x, y) = \frac{2}{y^3} \begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix} = \frac{2}{y^3} \begin{pmatrix} y \\ -x \end{pmatrix} \begin{pmatrix} y \\ -x \end{pmatrix}^\top \succcurlyeq 0,$$

since the right factor is rank-one PSD. Hence f is convex on the open half-plane (Figure 1.8).

- **Log-sum-exp** $f(\mathbf{x}) = \log \sum_{k=1}^n e^{x_k}$. Writing $z_k = e^{x_k} > 0$ and $\mathbf{1} = (1, \dots, 1)^\top$, a direct computation gives

$$\nabla^2 f(\mathbf{x}) = \frac{1}{\mathbf{1}^\top \mathbf{z}} \text{diag}(\mathbf{z}) - \frac{1}{(\mathbf{1}^\top \mathbf{z})^2} \mathbf{z} \mathbf{z}^\top.$$

For any $\mathbf{v} \in \mathbb{R}^n$,

$$\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} = \frac{(\sum_k z_k v_k^2)(\sum_k z_k) - (\sum_k z_k v_k)^2}{(\sum_k z_k)^2}.$$

Apply Cauchy–Schwarz to the vectors with components $a_k = \sqrt{z_k} v_k$ and $b_k = \sqrt{z_k}$:

$$\left(\sum_k a_k b_k \right)^2 \leq \left(\sum_k a_k^2 \right) \left(\sum_k b_k^2 \right) \iff \left(\sum_k z_k v_k \right)^2 \leq \left(\sum_k z_k v_k^2 \right) \left(\sum_k z_k \right).$$

The numerator above is therefore ≥ 0 , so $\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} \geq 0$ for every $\mathbf{v}$: $\nabla^2 f \succeq 0$. The same proof reappears in Unit 3 to certify the convexity of the multinomial logistic loss.

9.10 Operations that preserve convexity

The most efficient route to convexity for a complicated $L(\theta)$ is rarely Hessian computation; more often, L is written as a composition of operations known to preserve convexity.

Theorem 1.31 (Preservation of convexity). Let $f, g, f_1, \dots, f_m$ be convex functions on $\mathbb{R}^n$.

1. **Nonnegative combination.** For $\alpha_1, \dots, \alpha_m \geq 0$, the function $\alpha_1 f_1 + \dots + \alpha_m f_m$ is convex.
2. **Pointwise maximum / supremum.** $\max\{f_1, \dots, f_m\}$ is convex; for any index set $\mathcal{A}$ with each f_a convex, $\sup_{a \in \mathcal{A}} f_a(\mathbf{x})$ is convex.
3. **Affine composition.** If f is convex, so is $\mathbf{x} \mapsto f(\mathbf{A}\mathbf{x} + \mathbf{b})$ for any matrix $\mathbf{A}$ and vector $\mathbf{b}$.

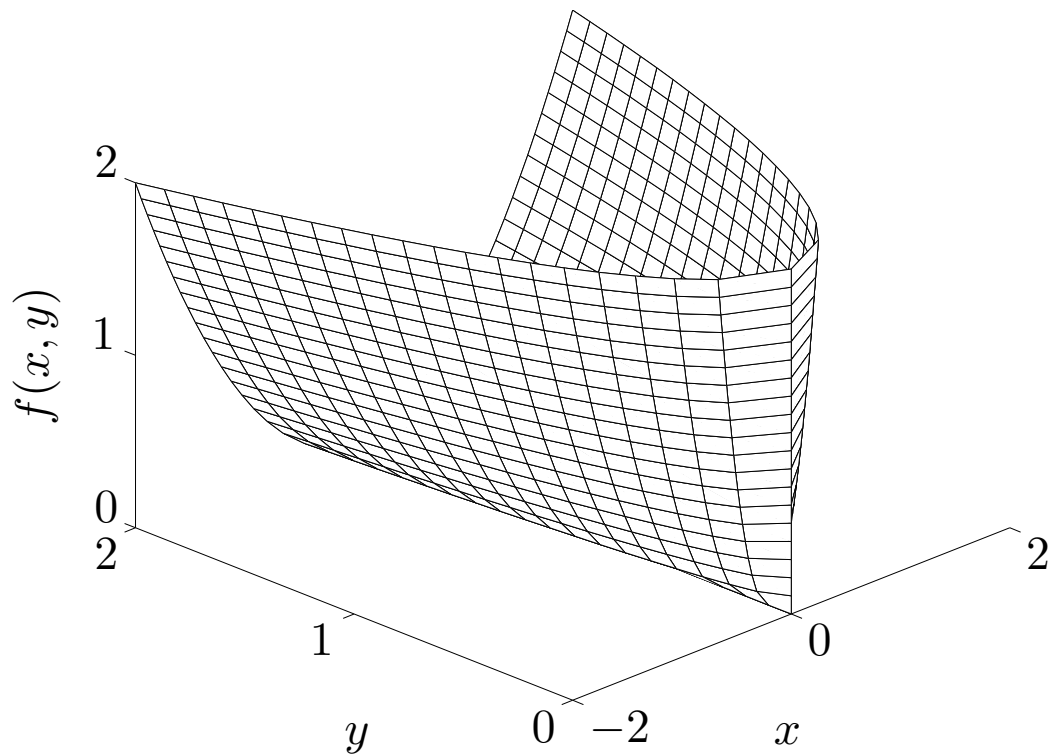


Figure 1.8: Graph of the quadratic-over-linear function $f(x, y) = x^2/y$ on the half-plane $y > 0$.

Proof of Theorem 1.31

Throughout, let $\mathbf{x}, \mathbf{y}$ be in the relevant domain and $\theta \in [0, 1]$.

(1) Nonnegative combination. Let $F = \sum_i \alpha_i f_i$ with $\alpha_i \geq 0$. Then

$$\begin{aligned} F(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) &= \sum_i \alpha_i f_i(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) \\ &\leq \sum_i \alpha_i [\theta f_i(\mathbf{x}) + (1 - \theta) f_i(\mathbf{y})] \\ &= \theta \sum_i \alpha_i f_i(\mathbf{x}) + (1 - \theta) \sum_i \alpha_i f_i(\mathbf{y}) = \theta F(\mathbf{x}) + (1 - \theta) F(\mathbf{y}), \end{aligned}$$

where the inequality is convexity of each f_i multiplied by $\alpha_i \geq 0$ and summed.

(2) Pointwise maximum / supremum. Set $F(\mathbf{x}) = \sup_{a \in \mathcal{A}} f_a(\mathbf{x})$. For each fixed a ,

$$f_a(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) \leq \theta f_a(\mathbf{x}) + (1 - \theta) f_a(\mathbf{y}) \leq \theta F(\mathbf{x}) + (1 - \theta) F(\mathbf{y}),$$

where the first step is convexity of f_a and the second uses $f_a(\mathbf{x}) \leq F(\mathbf{x})$ and $f_a(\mathbf{y}) \leq F(\mathbf{y})$ with $\theta, 1 - \theta \geq 0$. The right-hand side does not depend on a ; taking the supremum over a on the

left gives $F(\theta\mathbf{x} + (1 - \theta)\mathbf{y}) \leq \theta F(\mathbf{x}) + (1 - \theta)F(\mathbf{y})$. The pointwise maximum is the special case $\mathcal{A} = \{1, \dots, m\}$.

(3) Affine composition. Let $h(\mathbf{x}) = f(\mathbf{Ax} + \mathbf{b})$. Linearity of the affine map gives

$$\mathbf{A}(\theta\mathbf{x} + (1 - \theta)\mathbf{y}) + \mathbf{b} = \theta(\mathbf{Ax} + \mathbf{b}) + (1 - \theta)(\mathbf{Ay} + \mathbf{b}),$$

so

$$\begin{aligned} h(\theta\mathbf{x} + (1 - \theta)\mathbf{y}) &= f(\theta(\mathbf{Ax} + \mathbf{b}) + (1 - \theta)(\mathbf{Ay} + \mathbf{b})) \\ &\leq \theta f(\mathbf{Ax} + \mathbf{b}) + (1 - \theta)f(\mathbf{Ay} + \mathbf{b}) = \theta h(\mathbf{x}) + (1 - \theta)h(\mathbf{y}). \end{aligned}$$

The inequality is convexity of f . The domain $\text{dom } h = \{\mathbf{x} : \mathbf{Ax} + \mathbf{b} \in \text{dom } f\}$ is the inverse image of a convex set under an affine map and is therefore convex. ■

Example 1.5. (*Why the Lasso objective is convex.*) The Lasso loss is

$$L(\boldsymbol{\beta}) = \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1, \quad \lambda \geq 0.$$

The first term is least-squares (convex by Theorem 1.30); the second is a norm (convex by Theorem 1.19); their nonnegative combination is convex by Theorem 1.31(1). Unit 2 will exploit this convexity to derive the soft-thresholding update via a (sub)gradient version of Theorem 1.23.

Further operations

Three additional convexity-preserving operations appear less frequently: *partial minimization* ($\inf_{\mathbf{y}} h(\mathbf{x}, \mathbf{y})$ is convex when h is jointly convex), *perspective* ($(\mathbf{x}, t) \mapsto tf(\mathbf{x}/t)$ for $t > 0$), and *scalar composition with a convex non-decreasing outer function*. The complete catalogue is in Boyd & Vandenberghe, Ch. 3.2.

9.11 Jensen's inequality

Theorem 1.32 (Jensen). Let f be convex and $\mathbf{Z}$ a random vector with $\mathbf{P}\{\mathbf{Z} \in \text{dom } f\} = 1$ and $\mathbf{E}[\mathbf{Z}]$ finite. Then

$$f(\mathbf{E}[\mathbf{Z}]) \leq \mathbf{E}[f(\mathbf{Z})].$$

The basic chord inequality $f(\theta\mathbf{x} + (1 - \theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta)f(\mathbf{y})$ is the special case where $\mathbf{Z}$ takes

only two values $(\mathbf{x}, \mathbf{y})$ with probabilities $(\theta, 1 - \theta)$. Jensen extends this to arbitrary distributions. The proof uses one preliminary fact: the expectation operator is linear on affine functions, i.e. $\mathbf{E}[\mathbf{a}^\top \mathbf{Z} + b] = \mathbf{a}^\top \mathbf{E}[\mathbf{Z}] + b$ for any deterministic $\mathbf{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}$.

Proof of Theorem 1.32

Set $\mathbf{x}_0 = \mathbf{E}[\mathbf{Z}]$. We assume $\mathbf{x}_0$ lies in the interior of $\text{dom } f$, which holds provided the distribution of $\mathbf{Z}$ is not concentrated entirely on the boundary of $\text{dom } f$; the boundary case requires the relative-interior subgradient theory of Rockafellar and is not pursued here. By Theorem 1.29, $\partial f(\mathbf{x}_0)$ is non-empty; choose any $\mathbf{g} \in \partial f(\mathbf{x}_0)$. (For differentiable f one may take $\mathbf{g} = \nabla f(\mathbf{x}_0)$.) The subgradient inequality (Definition 1.24) gives

$$f(\mathbf{z}) \geq f(\mathbf{x}_0) + \mathbf{g}^\top (\mathbf{z} - \mathbf{x}_0), \quad \forall \mathbf{z} \in \text{dom } f.$$

Substitute $\mathbf{z} = \mathbf{Z}$ (which lies in $\text{dom } f$ almost surely by hypothesis) and take expectations on both sides; by linearity of $\mathbf{E}[\cdot]$ on the affine right-hand side,

$$\mathbf{E}[f(\mathbf{Z})] \geq f(\mathbf{x}_0) + \mathbf{g}^\top \mathbf{E}[\mathbf{Z} - \mathbf{x}_0] = f(\mathbf{E}[\mathbf{Z}]),$$

since $\mathbf{E}[\mathbf{Z} - \mathbf{E}[\mathbf{Z}]] = \mathbf{0}$. ■

Two applications you will see.

- **Moment-generating function bound.** For $Z \sim \mathbf{N}(\mu, \sigma^2)$ and the convex $f(z) = e^z$, Jensen gives $\mathbf{E}[e^Z] \geq e^\mu$. The exact value $\mathbf{E}[e^Z] = e^{\mu + \sigma^2/2}$ shows the inequality is strict when $\sigma > 0$.
- **Information-theoretic decomposition.** In Unit 3, Jensen's inequality decomposes cross-entropy loss into the entropy of the true class distribution and the KL divergence between predicted and true distributions, demonstrating equivalence between cross-entropy minimization and KL minimization.

9.12 Why convexity matters

Theorem 1.33. Let f be convex on a convex set C . Then every local minimizer of f on C is a global minimizer. If f is strictly convex, the minimizer, when it exists, is unique.

Proof of Theorem 1.33

Suppose $\mathbf{x}^* \in C$ is a local but not global minimizer, so there exists $\mathbf{y} \in C$ with $f(\mathbf{y}) < f(\mathbf{x}^*)$. For $\theta \in (0, 1]$, set $\mathbf{z}_\theta = (1 - \theta)\mathbf{x}^* + \theta\mathbf{y} \in C$ (using convexity of C). By convexity of f ,

$$f(\mathbf{z}_\theta) \leq (1 - \theta)f(\mathbf{x}^*) + \theta f(\mathbf{y}) < f(\mathbf{x}^*),$$

since $f(\mathbf{y}) < f(\mathbf{x}^*)$ and $\theta > 0$. As $\theta \rightarrow 0^+$, $\mathbf{z}_\theta \rightarrow \mathbf{x}^*$, so every neighbourhood of $\mathbf{x}^*$ contains a point with strictly lower f , contradicting that $\mathbf{x}^*$ is a local minimizer. Hence every local minimizer is global.

For uniqueness under strict convexity, suppose $\mathbf{x}^* \neq \mathbf{y}^*$ are both global minimizers with $f(\mathbf{x}^*) = f(\mathbf{y}^*) = f^*$. The midpoint $\mathbf{z} = \frac{1}{2}\mathbf{x}^* + \frac{1}{2}\mathbf{y}^* \in C$ satisfies $f(\mathbf{z}) < \frac{1}{2}f(\mathbf{x}^*) + \frac{1}{2}f(\mathbf{y}^*) = f^*$ by strict convexity, contradicting minimality of f^* . ■

Practical consequence: when $L(\theta)$ is convex, gradient descent, coordinate descent, and Newton's method converge to the unique optimum, with no risk of being trapped at a suboptimal local minimum. This is why most algorithms in the first half of the course are formulated as convex problems.

Forward links

Unit 2 (Ridge, Lasso, Elastic Net) and Unit 3 (logistic regression) operate entirely within Theorem 1.33. Unit 6 (SVM) is convex quadratic and uses Lagrangian duality. Tree-based methods (Unit 5) and K-means (Unit 7) depart from this setting: tree fitting is a greedy combinatorial procedure with no convex objective, and K-means alternates between two convex sub-problems while the joint problem is non-convex. These features are addressed in their respective units.

10 Pitfalls and Misapplications

The failure modes of §§ 1.1–1.9, each accompanied by a short R demonstration. These pitfalls form the trap-question pool for the final project.

10.1 Pitfall 1: Modeling without EDA

The proximate cause of most other pitfalls on this list. A brief `glimpse()` catches column-type and scale errors and obvious leakage candidates that no model can subsequently recover. The remedy is procedural: a structured EDA pass precedes every `fit()` call (§3.5).

10.2 Pitfall 2: Identifier columns as numeric features

R imposes no type constraint on input columns. A linear model fits a coefficient on `customer_id`; a tree splits on it. Both report training scores that lack predictive content in deployment, where the identifier values are entirely different.

```
1 set.seed(1); n <- 200
2 df <- tibble(
3   customer_id = 1001:(1000 + n),
4   age         = rnorm(n, 40, 10),
5   churned     = rbinom(n, 1, plogis(-0.05 * (rnorm(n, 40, 10) - 40)))
6 )
7 fit <- glm(churned ~ customer_id + age, data = df, family = binomial)
8 round(coef(fit), 6)
```

```
## (Intercept) customer_id      age
##      1.091192   -0.000600   -0.019256
```

The non-zero coefficient on `customer_id` indicates the model is treating identifier values as signal. The column should be dropped before fitting, or removed at recipe level via `step_rm(customer_id)` in Unit 2.

10.3 Pitfall 3: Post-event features

A column populated only after y is observed is unavailable at prediction time. §7.3 provides a runnable demonstration; the standing remedy is the timing constraint $t_{\text{rec}} \leq t_{\star}$ for every feature, enforced as a column-by-column audit (§4.1).

10.4 Pitfall 4: Accuracy on imbalanced data

On a 99/1 split, the majority-class predictor achieves 99% accuracy with zero recall. §8.3 provides the runnable demonstration; Unit 3 introduces precision, recall, and threshold tools that align the metric with the actual cost of error.

10.5 Pitfall 5: Mean imputation as a default

Mean imputation is unbiased only under MCAR. Under MAR it biases coefficient estimates; under MNAR it both biases the estimates and discards the signal carried by the missingness pattern. The recommended default is to encode the missingness indicator before imputing values.

```
1 set.seed(1); n <- 500
2 x           <- rnorm(n)
3 miss        <- rbinom(n, 1, 0.3)
4 y           <- ifelse(miss == 1, 1, 0)           # MNAR: y *is* the missingness pattern
```

```
5 x_obs      <- ifelse(miss == 1, NA, x)
6
7 # (a) Naive mean-impute: signal lost.
8 x_mean     <- ifelse(is.na(x_obs), mean(x_obs, na.rm = TRUE), x_obs)
9 cor(x_mean, y)
```

```
## [1] -1.968071e-21
```

```
1 # (b) Encode missingness as a feature, then impute: signal recovered.
2 miss_flag <- as.integer(is.na(x_obs))
3 cor(miss_flag, y)
```

```
## [1] 1
```

The naive correlation is near zero; the missingness indicator carries the full signal in y .

10.6 Pitfall 6: Random split on time-series data

Data with a temporal axis must be split chronologically: training on the past, validation on the future. A random split places future rows in the training set and requires the model to predict past rows, inverting the deployment direction. Unit 4 replaces random K-fold with rolling-origin and grouped variants for this reason.